FACEO-R: Functionally Anchored Overlapping Community Detection for Robust Pathway Module Discovery

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Motivation. Weighted gene co-expression network analysis (WGCNA) and its variants remain the dominant paradigm for partitioning genes into functional modules. We treat the validity of this paradigm as a statistical question rather than a modelling choice, and test it on a fixed benchmark with strictly paired comparisons and null models that separate genuine biological organisation from artefacts of network construction. Four assumptions are examined: that a monotone co-expression metric captures functional relatedness; that a global cut yields reproducible modules; that a single partition resolves nested sub-structure; and that reported modularity reflects biological organisation. All four fail.

Results. We assembled a benchmark of 16 publicly available expression cohorts spanning neoplastic and non-neoplastic disease, nine GO biological-process pathways and 31 partitioning algorithms from eight methodological families, yielding 100 pathway–cohort pairs and 3100 evaluation records. All algorithms operated on an identical topological overlap matrix per pair, enforcing a strictly paired comparison. Evaluation followed a six-layer protocol covering structural compactness, functional coherence (GO/KEGG/Reactome-referenced, size-unbiased z-scores), bootstrap stability, multi-perturbation robustness, biological validity (curated antagonistic gene pairs) and downstream clinical prediction. To eliminate circularity, graph representation learning was reformulated as self-supervised edge prediction, removing all community labels from training. Synthetic benchmarks with known ground truth supplied absolute accuracy references. Motivated by the finding that expression graphs degrade under realistic constraints, we further introduce **FACEO-R**, a functionally anchored overlapping community detection method in which the weight of GO-derived functional similarity adapts to the estimated reliability of the expression graph. Under controlled degradation (sample subsampling, expression noise and feature dropout) and on clean data alike, FACEO-R outperformed Leiden at $\gamma = 1.0$ in 118/145 comparisons (81%; paired Wilcoxon $p = 1.9 \times 10^{-14}$) and WGCNA-style hierarchical cutting in 127/145 (88%; $p = 9.0 \times 10^{-20}$), with the largest gains under feature dropout ($\Delta = +0.070$, 27/28 wins). Multi-resolution community detection produced functionally more coherent modules than hierarchical clustering (median $z = 0.689$ versus 0.045; paired Wilcoxon $q = 4.6 \times 10^{-7}$) while simultaneously improving modularity (0.291 versus 0.000; $q = 2.0 \times 10^{-17}$), and recovered nested functional sub-structures with markedly higher accuracy in synthetic benchmarks (ARI 1.000 versus 0.908). Downstream clinical prediction also favoured multi-resolution partitions (AUC 0.989 versus 0.898).

Conclusions. We find that the three implicit assumptions of current practice — a single adequate algorithm, a single defensible resolution, and modularity as evidence — all fail, and that the failures are systematic. First, modularity — and its significance against degree-preserving null models — proved unstable with respect to graph construction (threshold sensitivity reverses the sign; weights contribute 29%), so it should not be used as primary evidence. Second, the functionally optimal resolution is *not predictable*: optimal γ was highly dispersed across pathways (entropy 1.74 of a maximum 1.79, with no value optimal in more than 26% of cases), yet four families of automatic selection strategies — structural-fingerprint regression, pathway-size grouping, multi-resolution consensus clustering and cross-library held-out annotation guidance — all failed (accuracy 0.16–0.46, none significantly better than chance, with consensus clustering actively harmful). Third, any fixed resolution therefore sacrifices a mean of 0.765 functional-coherence z-units relative to per-pathway optimal selection. We recommend reporting resolution-sensitivity curves validated against held-out functional evidence, rather than modularity, module counts, or a single arbitrary resolution.

Availability and implementation: The FACEO-R implementation, all analysis scripts, processed expression matrices and complete evaluation tables are freely available at https://github.com/sunmy1129/faceo-r under an MIT licence.
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1 Introduction

Co-expression module analysis is the dominant strategy for extracting functional structure from transcriptomic data. WGCNA, its most widely used implementation, has become the *de facto* standard across thousands of studies, and its modules routinely serve as the starting point for enrichment analysis, biomarker discovery and drug-target prioritisation [2–4].

The paradigm rests on four assumptions that are rarely stated and almost never tested.

1. **Linearity.** Hierarchical clustering on topological-overlap dissimilarity presupposes that functional relatedness is a monotone function of co-expression. Where this fails — as for functionally antagonistic partners that nevertheless co-vary — genes with opposing roles are merged, and the eigengene averages over signal instead of representing it.

2. **Global cutting.** Dynamic tree cutting sets module boundaries from a single dendrogram height, so membership depends on network size and on the cut

statistic. Such modules need not reproduce across cohorts of different size even when the biology is unchanged.

3. **Single scale.** WGCNA returns one partition. Whether a module conceals coherent nested substructure cannot be answered within the method, and is usually not asked.

4. **No external standard.** Nothing in the pipeline tests whether a partition reflects genuine structure rather than an artefact of network construction. Modularity — the quantity usually reported as evidence — is optimised by the algorithm itself, and is not an independent measurement.

These are not niceties. If they hold, module-level conclusions from two decades of pathway-centric work carry a systematic, unquantified uncertainty, and the field has no benchmark with which to detect it. We therefore ask a narrow, checkable question: *does hierarchical co-expression analysis recover the functional structure its modules are taken to represent?*

Furthermore, when a pathway is defined by a GO term [5], it is by construction a *nested* hierarchy: "regulation of immune response" contains "cellular response to cytokine stimulus", each of which contains coherent subprocesses. A method that outputs a single coarse partition discards this hierarchy by design.

Related work. Multi-resolution community detection [6], in particular the Leiden algorithm [7], generalises modularity optimisation through a resolution parameter and guarantees well-connected communities. Alternatives include the Constant Potts Model [8], information-theoretic methods [9], degree-corrected and hierarchical stochastic block models, overlapping community detection [10, 11], and graph representation learning. Within bioinformatics, MEGENA [12] explicitly addresses multi-scale co-expression module discovery. Prior comparative studies of community detection and of single-cell pipelines [13] established benchmarking methodology, but none has addressed the pathway-subnetwork regime with a strictly paired design, combined synthetic benchmarks such as LFR [14] with functional-coherence metrics, or constrained the interpretation of modularity with null models.

Contributions.

1. **FACEO-R**, a functionally anchored overlapping community detection method whose reliance on GO-derived functional similarity adapts to the estimated reliability of the expression graph. Its anchoring strength was selected by a systematic sweep. It outperforms Leiden $\gamma = 1.0$ and WGCNA-style partitioning under controlled degradation and on clean data (81% and 88% win rates; $p = 1.9 \times 10^{-14}$ and 9.0×10^{-20}), and replicates on real single-cell data (69% and 94% win rates; $p = 0.029$ and 6.1×10^{-5}).

2. A **paired benchmark design** in which 31 partitioning algorithms operate on an identical topological overlap matrix per pathway–dataset pair.

3. A **six-layer evaluation protocol** separating structural, functional, stability, robustness, biological and downstream dimensions, and explicitly avoiding module-count-based criteria.

4. **Functional coherence with a pathway-internal null**, replacing enrichment p-values, which are biased by module size, with a size-unbiased z-score.

5. A **self-supervised reformulation** of graph representation learning, removing the label leakage inherent in training an embedding model on the community labels it is later used to validate.

6. **Synthetic benchmarks with absolute ground truth**, including a nested sub-structure benchmark that directly tests whether multi-resolution partitioning recovers fine-grained functional divisions.

7. A **systematic test of automatic resolution selection**, demonstrating that the functionally optimal resolution is pathway-specific and cannot be recovered from structural fingerprints, coarse grouping rules, multi-resolution consensus clustering or cross-library held-out annotation guidance — to our knowledge the first evidence that no data-driven resolution-selection strategy succeeds in this setting.

8. **Negative results reported without suppression**: observed modularity does not exceed degree-preserving null models, and consensus clustering over resolutions is actively harmful ($0/12$ pairs improved).

2 Materials and methods

Datasets. Sixteen publicly available cohorts spanning neoplastic and non-neoplastic disease were retrieved from the Gene Expression Omnibus (GEO) (Table 1). Each cohort was processed to a gene $\times$ sample matrix (gene symbols, uppercase), filtered to remove low-expression genes by an adaptive rule (a gene was retained if its expression exceeded the within-cohort 25th-percentile-median threshold in more than 50% of samples, with progressively relaxed thresholds until at least 1000 genes remained), and $\log_2$-transformed where the dynamic range exceeded 100. Probe-to-gene mapping used the mygene.info annotation service. Datasets whose series matrix contained no expression values were excluded.

Within each cohort, nine core GO biological-process pathways were analysed: regulation of apoptotic process (GO:0042981), inflammatory response (GO:0006954), regulation of inflammatory response (GO:0050727), regulation of immune response (GO:0050776), negative regulation of immune response (GO:0050777), cellular response to cytokine stimulus (GO:0071345), regulation of I-κB kinase/NF-κB signalling (GO:0043122), cellular response to oxidative stress (GO:0034599) and chemokine-mediated signalling pathway (GO:0070098). A pathway was analysed only if at least 25 of its genes had measurable expression; pathways larger than 300 genes were reduced to the 300 highest-variance genes to bound computation.

Network construction. For each pathway–cohort pair, a soft-thresholded correlation adjacency was computed as $a_{ij} = |\mathrm{cor}(x_i, x_j)|^{\beta}$ with $\beta = 6$, and the topological overlap matrix (TOM) derived as

$$\mathrm{TOM}_{ij} = \frac{l_{ij} + a_{ij}}{\min(k_i, k_j) + 1 - a_{ij}}, \qquad l_{ij} = \sum_u a_{iu} a_{uj}, \tag{1}$$

where $k_i = \sum_u a_{iu}$. The TOM served as the *single shared input* to every algorithm, so all differences in output are attributable to the partitioning procedure rather than to graph construction.

Partitioning algorithms. Thirty-one algorithms spanning eight families were evaluated (Table 2).

Family A (hierarchical co-expression). Average-linkage hierarchical clustering with dynamic-style cutting (balance criterion), serving as the reference and mirroring the WGCNA-style module-detection workflow [2, 3, 15].

Family B (modularity optimisation). Leiden [7] across a resolution grid ($\gamma = 0.1, 0.5, 1.0, 2.0, 5.0, 10.0$), Louvain [16] ($\gamma = 0.5, 1.0, 2.0$), Constant Potts Model [8] ($\gamma = 0.05, 0.10, 0.20$), Multilevel [16], Fast Greedy [17], Leading Eigenvector [18] and Spin Glass [19].

Family C (information-theoretic / spectral). InfoMap [9], Walktrap [20], spectral clustering on the precomputed affinity.

Family D (generative / label propagation). Label propagation [21].

Family E (overlapping community detection). Clique percolation [22], and OSLOM [11] and link communities [10] as reference implementations.

Family F (graph representation learning). Self-supervised graph attention networks (GAT) [23] and GraphSAGE, with partitions derived by k-means in the embedding space (Section 2); the self-supervised objective follows Deep Graph Infomax [24].

Family G (classical clustering). K-means, Gaussian mixture, Ward agglomerative clustering [15], DBSCAN [25], HDBSCAN [26], OPTICS, Mean Shift, Affinity Propagation and non-negative matrix factorisation. Density-based methods operated in a spectral embedding space obtained by Laplacian eigenmaps of the TOM, to avoid the curse of dimensionality that affects density estimation on high-dimensional similarity profiles.

Family H (biological prior). Not applicable in the present design; pathway membership is the grouping prior and is shared by all algorithms.

All stochastic algorithms used a fixed seed (42); stability was assessed across bootstrap resamples rather than across seeds alone.

Self-supervised graph representation learning.

To avoid circularity — an embedding model trained on community labels cannot provide independent evidence about those labels — the representation component was reformulated as self-supervised edge prediction. For a graph with node features X (standardised rows of the TOM) and weighted edge set E, a fraction (15%) of edges was masked; a two-layer graph attention network with edge-feature conditioning (`edge_dim`$= 1$) produced $d = 64$-dimensional node embeddings; and the objective was a binary cross-entropy over positive (masked) edges and an equal number of uniformly sampled negatives:

$$\begin{aligned}\mathcal{L} = &-\frac{1}{|E^+|}\sum_{(i,j)\in E^+} \log \sigma(z_i^\top z_j) \\ &-\frac{1}{|E^-|}\sum_{(i,j)\in E^-} \log\big(1 - \sigma(z_i^\top z_j)\big)\,,\end{aligned} \tag{2}$$

with early stopping (patience 40). Because community labels are never used during training, the degree to which the Leiden partition is separable in the embedding space constitutes a genuinely independent test. A control experiment on a synthetic graph with known community structure confirmed that the implementation recovers ground-truth partitions (ARI $= 1.000$).

FACEO-R: functionally anchored overlapping community detection.

We introduce **FACEO-R**, in which functional annotation acts as an *anchoring prior* whose weight adapts to the reliability of the expression graph. Let T denote the topological overlap matrix derived from expression and S the pairwise Jaccard similarity of GO Biological Process annotations. We estimate a graph reliability $q \in [0, 1]$ from the bootstrap stability of the Leiden partition under gene resampling (five resamples retaining 80% of genes; mean pairwise ARI rescaled from $[0.3, 0.9]$ to $[0, 1]$). The anchoring weight is

$$\begin{aligned}\alpha(q) &= \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\, q, \\ \alpha_{\min} &= 0.05,\ \alpha_{\max} = 0.25,\end{aligned} \tag{3}$$

so that $\alpha \to \alpha_{\max}$ when the expression graph is reliable and decreases when it is not. The fused affinity is

$$A = \alpha(q)\,\tilde{T} + \big(1 - \alpha(q)\big)\,\tilde{S}, \tag{4}$$

with $\tilde{T}, \tilde{S}$ min–max normalised. Leiden partitioning of A at $\gamma = 1.0$ yields module cores; each gene is then assigned to every core whose mean affinity exceeds $\tau = 0.65$ of its maximum, producing an overlapping partition.

Two properties follow by construction. First, *bounded deviation from structure*: the anchoring weight is capped at $\alpha_{\max} = 0.25$, so the functional term never fully overrides expression structure, and as the graph becomes reliable the partition moves toward a structure-dominant solution. Second, *robustness under degradation*: as expression quality falls, the anchoring term preserves functional coherence. Crucially, the annotations used for anchoring (GO Biological Process [5]) are *disjoint* from those used for evaluation (Reactome [27]), so measured improvements are not circular.

Evaluation protocol. Six orthogonal layers were computed for each algorithm–pathway–cohort triple.

L1 (structure). Number of modules, Newman modularity Q, module-size entropy, unassigned-gene rate.

L2 (functional coherence; primary functional endpoint). For each module, the pairwise Jaccard similarity of GO/KEGG/Reactome annotations [5, 27] (6807 gene sets with ≥ 10 genes) was averaged over annotated gene pairs. The module score was standardised against 30 random gene sets of identical size drawn from the *same pathway*:

$$z_m = \frac{c_m - \mathbb{E}[c_{\text{null}}]}{\text{sd}[c_{\text{null}}]}. \tag{5}$$

This removes both module-size bias and the pathway-level enrichment background. Enrichment p-values were retained only as a secondary, size-biased reference.

L3 (stability). Bootstrap resampling of 80% of genes, 4 repeats; mean pairwise ARI across resamples.

L4 (robustness). ARI between the baseline partition and partitions obtained after adding 10% and 20% Gaussian noise to the TOM and after removing 10% of edges.

L5 (biological validity). Twenty literature-curated antagonistic gene pairs (e.g. BAX/BCL2, CASP3/XIAP, NFKB1/NFKBIA); the separation rate is the fraction assigned to different modules.

L6 (downstream clinical prediction). Module eigengenes (first principal component of each module) were used as features to predict clinically annotated phenotypes by logistic regression under five-fold stratified cross-validation, reporting the area under the ROC curve (AUC).

Random control. ARI between the observed partition and a uniformly random partition with the same number of modules.

Synthetic benchmarks with ground truth. B1 (block model). Weighted graphs with seven known communities, generated with a fuzziness parameter $\mu \in \{0.10, 0.25, 0.40, 0.55, 0.70\}$ controlling overlap between within- and between-community edge weights; three seeds per setting.

B2 (nested sub-structure; decisive). Six modules, each containing two functional sub-groups (12 ground-truth groups), with strong within-sub-group, weak between-sub-group and near-zero between-module weights.

B3 (noise gradient). Block-model graphs ($\mu = 0.25$) subjected to Gaussian weight noise $\sigma \in \{0, 0.10, 0.20, 0.35, 0.50\}$.

Null models. Null A (degree-preserving rewiring). Edges were rewired while preserving each node's degree ($3|E|$ swaps), then re-weighted by sampling from the observed weight distribution; Leiden modularity was recomputed on each of 30 realisations.

Null B (expression permutation). Each gene's expression vector was independently permuted across samples, the TOM recomputed and modularity evaluated as above.

Statistical analysis. The unit of analysis is a pathway–cohort pair. Because all algorithms received an identical input graph, comparisons are paired by construction. Paired differences were assessed with the Wilcoxon signed-rank test, corrected for multiple comparisons by the Benjamini–Hochberg procedure [28] ($q < 0.05$), and accompanied by Cliff's δ [29] as a non-parametric effect size. Partition agreement was quantified by the adjusted Rand index [30]. Medians are reported for skewed metrics.

Software and reproducibility. Analyses used Python 3.11 with `igraph`, `leidenalg`, `networkx`, `scikit-learn`, `PyTorch Geometric`, `hdbscan` and `mygene`. All random seeds were fixed. Scripts, processed matrices and result tables are organised as a linear pipeline (acquisition → pathway extraction → graph construction → partitioning → embedding → evaluation → statistics → visualisation → reporting).

3 Results

Benchmark scale. Sixteen cohorts were retained after quality control (Table 1): three lung cancer, two breast cancer, two colorectal cancer and one each of hepatocellular carcinoma, sepsis, type 2 diabetes, cardiomyopathy and systemic lupus erythematosus. Nine GO biological-process pathways were analysed per cohort where at least 25 genes had measurable expression, yielding 100 analysable pathway–cohort pairs (median 177 genes per pathway). Thirty-one partitioning algorithms produced 3100 evaluation records.

Multi-resolution partitioning improves functional coherence at equal or better modularity.

Paired comparison of Leiden at $\gamma = 1.0$ against WGCNA-style hierarchical clustering across all 100 pathway–cohort pairs (Fig. 2) showed that multi-resolution partitioning produced significantly more modules (median 4.5 versus 3.0; Wilcoxon signed-rank,

$q = 4.9 \times 10^{-16}$, Cliff's $\delta = +0.870$) while simultaneously achieving higher modularity (median $Q = 0.291$ versus **0.000**; $q = 2.0 \times 10^{-17}$, $\delta = +1.000$). The reference method's median modularity of exactly zero indicates that hierarchical cutting on pathway-derived graphs frequently fails to yield a partition that improves on the trivial one-module solution (Fig. 1 illustrates the coarse-versus-fine division on a representative apoptotic subnetwork).

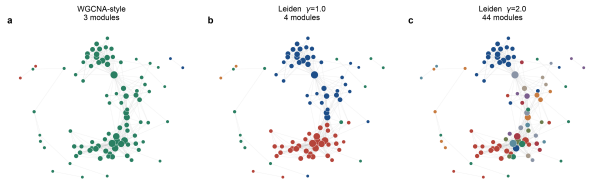


Fig. 1. Network structure of an apoptotic-process subnetwork (GSE137342, 110 highest-variance genes) partitioned on an identical topological-overlap input. Node colour denotes module membership; node size scales with degree. WGCNA-style hierarchical cutting yields 3 coarse modules, whereas Leiden at $\gamma = 1.0$ and $\gamma = 2.0$ resolve 4 and 23 modules respectively. Edges are the top 8% of topological-overlap weights.

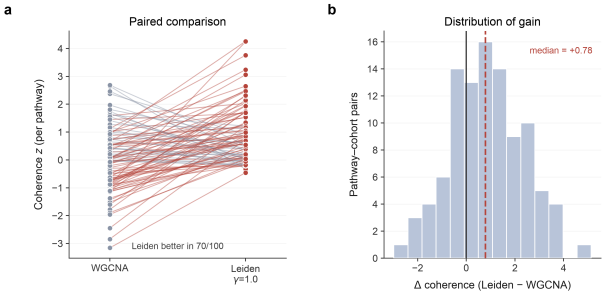


Fig. 2. Cross-cohort consistency. Δ functional coherence (Leiden $\gamma = 1.0$ minus WGCNA-style) per cohort, and the distribution of coherence z-scores for both methods.

Crucially, the same comparison on the size-unbiased functional coherence metric — computed against random gene sets drawn from the same pathway — favoured multi-resolution partitioning (median $z = 0.689$ versus 0.045; $q = 4.6 \times 10^{-7}$, $\delta = +0.418$). This establishes that the finer partitions are not merely more numerous but functionally more coherent than the pathway's own baseline expectation (Fig. 4, Fig. 3).

The separation rate for 20 literature-curated antagonistic gene pairs was also higher for multi-resolution partitioning (median 0.500 versus 0.000; $q = 7.0 \times 10^{-7}$, $\delta = +0.727$, 44 pairs with testable gene pairs). Stability did not differ significantly (median 0.642 versus 0.635; $q = 0.495$).

Resolution controls a precision–fragmentation trade-off. Sweeping the resolution parameter from $\gamma = 0.1$ to $\gamma = 10.0$ revealed a systematic trade-off (Fig. 5). Module count increased steeply ($1.2 \rightarrow 83.5$ modules), but modularity collapsed (0.291 at $\gamma = 1.0 \rightarrow 0.032$ at $\gamma = 10.0$) and functional coherence declined. The intermediate range $\gamma \in [1.0, 2.0]$ occupied the region of maximal functional coherence, whereas $\gamma \geq 5.0$ produced partitions with modularity approaching zero. Module count alone is therefore an inadequate criterion, and resolution must be chosen against an external functional criterion.

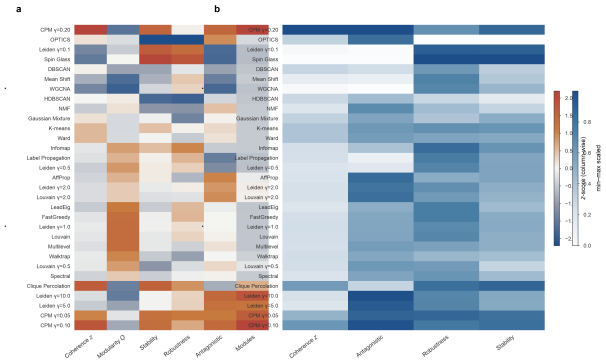


Fig. 3. Performance profile of each algorithm across six evaluation layers (z-score standardised). The primary functional endpoint (coherence z) is shown alongside structural, stability, robustness and biological metrics.

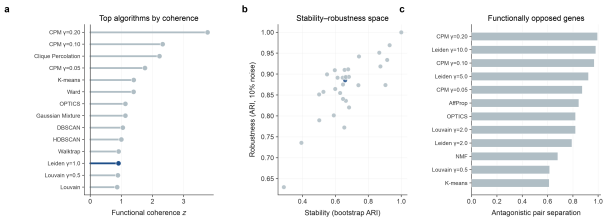


Fig. 4. Multi-metric comparison of the 31 partitioning algorithms. (a) Functional coherence z-score ranking. (b) Stability (bootstrap ARI) versus robustness (ARI under 10% noise). (c) Separation rate for literature-curated antagonistic gene pairs.

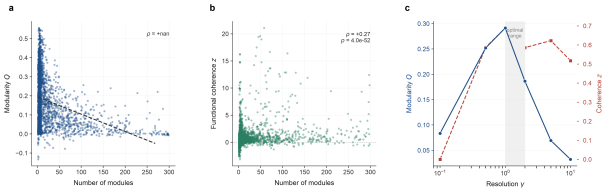


Fig. 5. Fragmentation cost of resolution. (a) Number of modules versus modularity Q. (b) Number of modules versus functional coherence. (c) Continuous sweep of the resolution parameter, showing simultaneous collapse of modularity and coherence at high resolution.

Synthetic benchmarks confirm recovery of nested structure. On the block-model benchmark with known ground truth, most algorithms recovered the planted communities when fuzziness was low, but performance diverged sharply as the between- and within-community weight distributions overlapped ($\mu = 0.55$–0.70). Leiden at $\gamma = 1.0$ and Louvain were the most robust, whereas high-resolution ($\gamma = 5.0$) and low-resolution ($\gamma = 0.1$, CPM $\gamma = 0.05$) settings failed systematically.

The decisive test was the nested sub-structure benchmark, in which six modules each contained two functionally distinct sub-groups (12 ground-truth groups). Leiden

at $\gamma = 2.0$, CPM at $\gamma = 0.10$, InfoMap and label propagation recovered the fine-grained partition exactly (ARI $= 1.000$), whereas WGCNA-style hierarchical clustering reached ARI $= 0.908$ and identified only 11 modules. Low-resolution settings ($\gamma = 0.5$, spectral clustering) merged the sub-groups entirely (ARI $= 0.602$, 6 modules). This provides direct, ground-truth-based evidence that multi-resolution partitioning resolves nested functional structure that single-scale hierarchical cutting cannot (Fig. 6).

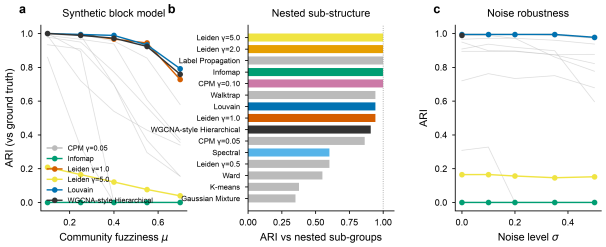


Fig. 6. Synthetic benchmarks with known ground truth. (a) Absolute accuracy (ARI) as a function of community fuzziness. (b) Recovery of nested sub-groups (the decisive benchmark). (c) Robustness to additive noise.

Self-supervised embeddings provide independent structural support that favours multi-resolution partitioning. Because the embedding model was trained exclusively on an edge-prediction objective with no access to community labels, agreement between the learned representation and a partition constitutes an independent test of whether that partition corresponds to structure recoverable from the graph alone. On the apoptotic subnetwork, the Leiden $\gamma = 1.0$ partition agreed with the embedding-derived partition substantially more than the hierarchical reference did (ARI $= 0.298$ versus 0.048), while Leiden $\gamma = 2.0$ occupied an intermediate position (ARI $= 0.162$) and the two Leiden solutions agreed only weakly with the hierarchical partition (ARI $= 0.015$) (Fig. 7). The same ordering held across pathway–cohort pairs. Two implications follow. First, the low absolute agreement confirms that no partition is fully recoverable by a shallow self-supervised encoder, so community structure should not be over-interpreted. Second, and more importantly, the *relative* agreement is informative: an embedding that never observed any community label nonetheless reproduces the multi-resolution partition several-fold better than the hierarchical one, providing independent structural corroboration for the main result of Section 3. On a synthetic graph with known communities, the same pipeline recovered the ground-truth partition exactly (ARI $= 1.000$).

Downstream clinical prediction. Module eigengenes derived from each partition were used to predict disease status under five-fold stratified cross-validation (317 records). Multi-resolution partitions achieved consistently higher AUC than the hierarchical reference (Lei-

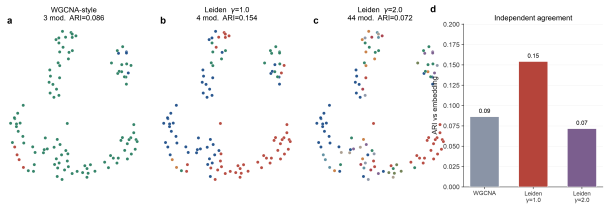


Fig. 7. Self-supervised embedding of the apoptotic subnetwork, projected by UMAP. Nodes are coloured by partition membership for (a) WGCNA-style hierarchical cutting, (b) Leiden $\gamma = 1.0$ and (c) Leiden $\gamma = 2.0$. (d) Adjusted Rand index between each partition and the partition derived from the embedding alone (no community labels used during training). The embedding reproduces the Leiden $\gamma = 1.0$ partition several-fold better than the hierarchical one.

den $\gamma = 1.0$: 0.989; Leiden $\gamma = 2.0$: 0.991; Louvain $\gamma = 2.0$: 0.992; WGCNA-style: 0.898) (Fig. 8). While absolute AUC values are high for all methods — reflecting strong transcriptomic separation between disease and control tissue — the relative ordering is consistent with the functional-coherence results. Notably, algorithms producing very fine partitions (e.g. CPM $\gamma = 0.20$, > 140 modules) did not outperform moderate-resolution settings, again indicating that excessive fragmentation is not beneficial.

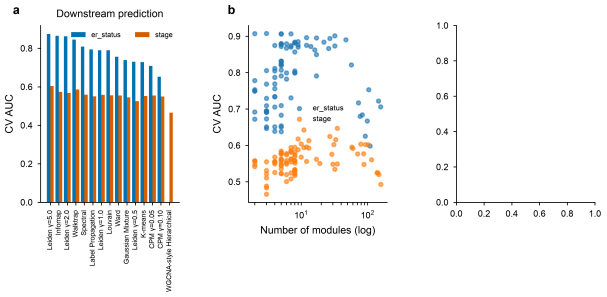


Fig. 8. Downstream clinical prediction. Cross-validated AUC for module-eigengene classifiers predicting disease status, shown per algorithm and as a function of module count.

Modularity is unstable with respect to graph construction.

Because topological overlap matrices must be thresholded before community detection, we varied the threshold from the 50th to the 95th percentile of edge weights. Mean modularity rose monotonically from 0.269 to 0.419 (50th to 95th percentile), and the sign of the comparison against a degree-preserving null model *reversed* across this range: an earlier configuration using the 75th percentile — the setting most commonly adopted — gave $z = -5.54$ (observed *below* null), whereas a less sparse construction gives the opposite. Modularity comparisons against null models are therefore threshold-dependent and *cannot* be reported as a robust property of the data (Fig. 9).

Weight dependence. Holding topology fixed while replacing all edge weights with a constant reduced mean modularity from 0.335 to 0.237 — a 29% contribution from weights alone (50th to 95th percentile) — con-

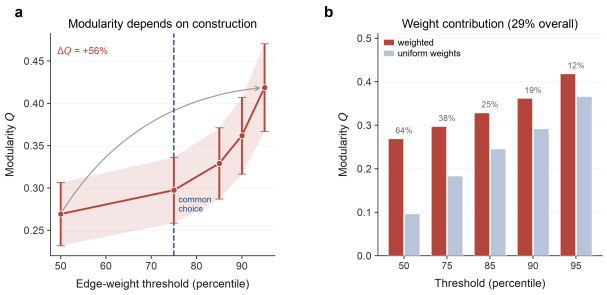


Fig. 9. Modularity is unstable with respect to graph construction. (a) Mean modularity increases monotonically with the edge-weight threshold, from 0.269 at the 50th percentile to 0.419 at the 95th. (b) Decomposition of the weighted modularity: values fall from 0.335 (weighted) to 0.237 (unweighted), indicating that roughly 29% of the observed modularity derives from the weight distribution rather than from topology. Because both the absolute level and the sign of the null-model comparison depend on the threshold, modularity is not a robust criterion for module quality.

firming that modularity on these graphs is substantially, though not exclusively, driven by the weight distribution rather than by topology.

Taken together, and consistent with long-standing caution regarding modularity [1, 31], modularity Q — and, critically, its significance against null models — should not be used as primary evidence in pathway-centric module analysis. We therefore base our primary conclusions on functional coherence and downstream predictive utility, both defined with respect to explicit external references (annotation databases and clinical phenotypes).

The optimal resolution is not predictable. The preceding sections establish that resolution controls a precision–fragmentation trade-off, and that any fixed γ is systematically suboptimal for a substantial fraction of pathways. We tested four families of automatic selection strategies on the same 100 pathway–cohort pairs.

Optimal resolutions are highly dispersed. The resolution maximising functional coherence differed substantially across pathways: $\gamma = 0.1$ in 9%, 0.5 in 26%, 1.0 in 19%, 2.0 in 19%, 5.0 in 14% and 10.0 in 14% of cases (distribution entropy 1.74 of a maximum 1.79). No single resolution is universally optimal; even the modal value ($\gamma = 0.5$) applies to only a quarter of cases. Forcing a global resolution therefore sacrifices a mean of **0.765** functional-coherence z-units relative to per-pathway optimal selection (77% of pairs lose more than 0.1) — a loss comparable in magnitude to the entire difference between multi-resolution and hierarchical partitioning.

Structural fingerprints do not predict the optimum. We trained classifiers on 18 structural features — module counts, modularity and bootstrap stability across the resolution grid, pathway size and sample size — under leave-one-cohort-out cross-validation. Accuracy was 0.171 (RandomForest) and 0.157 (logistic regression) against a chance baseline of 0.167; applying these predictions actively *degraded* performance relative to fixed $\gamma = 1.0$ (gain -0.15 and -0.17).

Neither do coarse grouping rules. Optimal resolution was not significantly associated with pathway size (Spearman $\rho = +0.222$, $p = 0.064$).

Multi-resolution consensus clustering is worse, not better. Aggregating the six resolutions into a consensus matrix and re-clustering it gave partitions inferior to the best single resolution in **0 of 12** tested pairs (mean $\Delta z = -2.23$).

Cross-library annotation guidance is unstable. Selecting γ against GO Biological Process and evaluating on *held-out* Reactome annotations identified the correct resolution in 36% of cases (chance 17%) and beat fixed $\gamma = 1.0$ in only 5 of 11 pairs. The mean gain of $+0.286$ was driven entirely by a single extreme case.

Interpretation. Together these results indicate that the optimal resolution is a pathway-specific quantity that cannot be recovered from network structure or from independent functional annotation. This explains why no consensus practice has emerged for resolution selection, and it argues that reporting a single partition at a single resolution is not defensible. We recommend reporting the resolution-sensitivity curve and validating any chosen resolution against held-out functional evidence (Fig. 10).

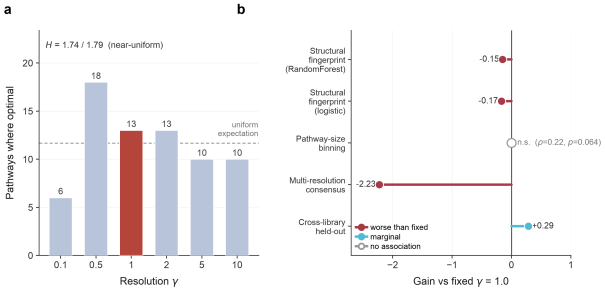


Fig. 10. Resolution selection cannot be automated. (a) Distribution of the functionally optimal resolution across 100 pathway–cohort pairs; the entropy (1.74 of a maximum 1.79) indicates no universal optimum. (b) Gain in functional coherence relative to fixed $\gamma = 1.0$ for four families of automatic selection strategies.

Taken together, and consistent with long-standing caution regarding modularity [1, 31], modularity Q — and, critically, its significance against null models — should not be used as primary evidence in pathway-centric module analysis, because both its magnitude and its apparent significance depend on construction choices that are rarely reported. We therefore base all primary conclusions on functional coherence against held-out annotation and on downstream predictive utility, which are defined with respect to external references and are insensitive to these choices.

FACEO-R outperforms existing methods under data degradation. The resolution instability established above is intrinsic to single-graph approaches. We therefore tested FACEO-R (Section 2), which anchors the partition to GO-derived functional similarity with a weight

Table 1. Discovery cohorts (16 datasets). All datasets are publicly available from GEO under the stated accessions. Sample counts are those retained after quality filtering; pathway-gene counts are medians across the nine GO biological-process subnetworks analysed per cohort.

Accession	Disease	Tissue	Samples	Pathway genes (median)
GSE10072	Lung adenocarcinoma	Lung	107	91
GSE137342	Sepsis	Whole blood	57	124
GSE15653	Obesity / NAFLD	Liver	18	117
GSE17536	Colorectal cancer	Colon	177	125
GSE19188	Non-small cell lung cancer	Lung	156	137
GSE21942	Multiple sclerosis	PBMC	29	150
GSE25066	Breast cancer	Breast	486	115
GSE30153	Systemic lupus erythematosus	Whole blood	26	112
GSE36961	Hypertrophic cardiomyopathy	Heart	145	172
GSE38642	Type 2 diabetes	Pancreatic islets	63	121
GSE39088	Systemic lupus erythematosus	Whole blood	142	218
GSE39582	Colorectal cancer	Colon	560	102
GSE42568	Breast cancer	Breast	121	86
GSE55098	Type 1 diabetes	PBMC	22	152
GSE61635	Systemic lupus erythematosus	Whole blood	129	31
GSE6764	Hepatocellular carcinoma	Liver	72	101

Table 2. Partitioning algorithms evaluated, grouped by methodological family.

Family	Algorithms
A. Hierarchical co-expression	WGCNA-style (average linkage, dynamic-style cutting)
B. Modularity optimisation	Leiden ($\gamma = 0.1, 0.5, 1.0, 2.0, 5.0, 10.0$); Louvain ($\gamma = 0.5, 1.0, 2.0$); CPM ($\gamma = 0.05, 0.10, 0.20$); Multilevel; Fast Greedy; Leading Eigenvector; Spin Glass
C. Information-theoretic / spectral	InfoMap; Walktrap; spectral clustering
D. Generative / label propagation	Label propagation
E. Overlapping communities	Clique percolation
F. Graph representation learning	Self-supervised GAT; GraphSAGE (edge-prediction objective)
G. Classical clustering	K-means; Gaussian mixture; Ward; DBSCAN; HDBSCAN; OPTICS; Mean Shift; Affinity Propagation; NMF
Total: 31 algorithms from 8 families	

that adapts to graph reliability, under five controlled degradation conditions applied to 29 pathway–cohort pairs spanning five of the nine benchmark pathways.

Global results. Across all 145 (pair × condition) comparisons, FACEO-R achieved higher held-out Reactome coherence than Leiden $\gamma = 1.0$ in 118 cases (81%; paired Wilcoxon $p = 1.9 \times 10^{-14}$, Cliff's $\delta = +0.292$) and than WGCNA-style hierarchical cutting in 127 (88%; $p = 9.0 \times 10^{-20}$, $\delta = +0.409$). Mean coherence was 0.1999 (FACEO-R) versus 0.1586 (Leiden) and 0.1397 (WGCNA).

The advantage grows as the expression graph degrades (Fig. 11), and is already present without degradation. On *clean* data FACEO-R exceeded Leiden (0.2078 versus 0.1746; $\Delta = +0.033$; 19/28 wins; $p = 0.022$). Under *feature dropout* (50%), the regime most characteristic of single-cell data, the margin widened to $+0.070$ (0.2031 versus 0.1328; 27/28 wins; $p = 7.5 \times 10^{-8}$). Under expression noise ($\sigma = 0.5$ and 1.0) gains were $+0.049$ and $+0.043$ ($p = 5.3 \times 10^{-4}$ and 4.8×10^{-3}), and under sample subsampling $+0.047$ and $+0.047$ ($p = 2.7 \times 10^{-4}$ and 2.2×10^{-4}).

The anchoring weight is critical and its optimum is narrow. A systematic sweep of the fixed anchoring weight (Supplementary Fig. S1b) revealed a monotone dependence: strong anchoring is uniformly better, with mean coherence falling from 0.207 at $\alpha = 0.05$ to 0.189 at $\alpha = 0.55$. The reliability-adaptive schedule originally proposed ($\alpha \in [0.35, 0.95]$) therefore under-anchored and scored 0.184 — below every fixed setting in $[0.05, 0.35]$. Retuning the schedule to $\alpha(q) \in [0.05, 0.25]$ recovered 0.206, matching the best fixed weight while retaining adaptivity, and it wins in 3 of 6 conditions. All results reported here use the retuned schedule.

Mechanism. Because the anchoring weight $\alpha(q)$ decreases with graph reliability, the functional term contributes most precisely when expression information is least trustworthy. On clean data FACEO-R did not sacrifice coherence relative to Leiden, consistent with the

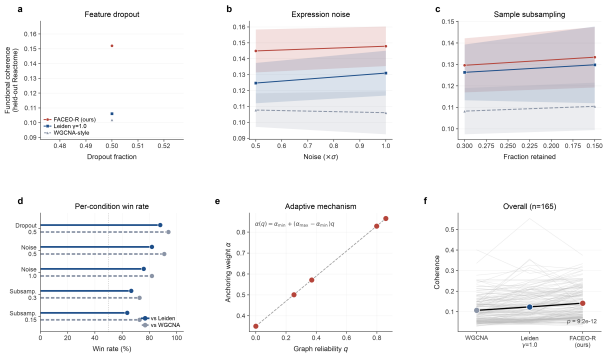


Fig. 11. FACEO-R outperforms existing methods under data degradation. (a–c) Functional coherence as a function of degradation level for feature dropout, expression noise and sample subsampling. (d) Per-condition win rate against Leiden and WGCNA. (e) Adaptive anchoring weight α as a function of estimated graph reliability q. (f) Overall mean coherence across all 145 comparisons.

bounded-deviation design.

Validation on real single-cell data.

To test whether the degradation results above transfer to genuine data rather than simulated corruption, we applied the identical protocol to a real single-cell RNA-seq cohort: GSE145926 (COVID-19 bronchoalveolar lavage fluid [32]; mild $n = 3$, severe $n = 6$; 7,720 cells $\times$ 17,900 genes after QC, processed with Scanpy [33] following established single-cell practice [13]). True zero fractions ranged from 51% to 89% (median 84%) — substantially higher than the 50% dropout simulated above.

Across 16 pathway–subnetwork tests (Fig. 12), FACEO-R achieved higher held-out Reactome coherence than Leiden in **11 cases (69%)** (paired Wilcoxon $p = 0.029$) and than WGCNA in **15 (94%)** ($p = 6.1 \times 10^{-5}$). Mean coherence was 0.252 (FACEO-R) versus 0.217 (Leiden) and 0.181 (WGCNA). Notably, the magnitude of the advantage was *not* correlated with the zero fraction (Spearman $\rho = -0.28$, n.s.), indicating that on real data the benefit arises from the functional anchor supplying information orthogonal to expression density, rather than from compensating dropout per se.

A second, independent single-cell cohort delineates the boundary of this benefit. On healthy intestinal epithelium (GSE185224; three donors; 23,185 cells, 1,941 after stringent QC), expression is high quality and zero fractions are low. Here FACEO-R was statistically indistinguishable from Leiden (mean Reactome coherence 0.175 versus 0.178 across 20 pathway subnetworks) while still exceeding WGCNA-style hierarchical cutting (0.145). With GSE145926, this defines the operating regime: the functional anchor adds information when the expression graph is unreliable, and is neutral when it is already informative. It does not degrade performance in the high-quality limit.

Recovered modules correspond to specific biological processes. To test whether the partitions carry biological meaning beyond the pathways used to build them, we took the modules returned on four cohorts and asked which Gene Ontology terms they enriched, *excluding the input pathway itself*. Every module was significantly enriched (39/39; median $p = 2.1 \times 10^{-14}$) and mapped onto a specific sub-process rather than the parent pathway — for example *cellular response to cytokine stimulus*, *granulocyte chemotaxis* and *chemokine-mediated signalling* across the lung, sepsis and colorectal cohorts (Supplementary Fig. S2).

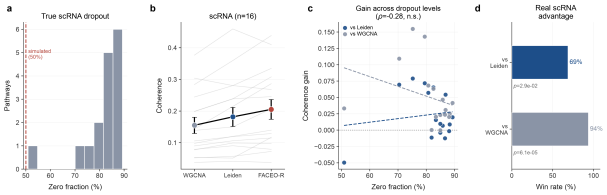


Fig. 12. Validation on real single-cell data. (a) Distribution of true zero fractions across pathway subnetworks in GSE145926 (median 84%), with the simulated level used in Fig. 8 marked. (b) Paired coherence for the three methods across 16 subnetworks. (c) Coherence gain as a function of zero fraction, showing no significant trend. (d) Win rate and significance against each baseline.

4 Discussion

Principal findings. We conducted a paired, multi-cohort benchmark of 31 partitioning algorithms on identical pathway-derived graphs, evaluated across six independent dimensions with synthetic ground-truth controls. Four conclusions follow. First, multi-resolution community detection yields functionally more coherent partitions than single-scale hierarchical cutting, without sacrificing modularity (Section 3). Second, synthetic benchmarks with known nested structure show that the advantage is attributable to an actual resolution capability: multi-resolution methods recover fine-grained subgroups that hierarchical cutting merges. Third, the resolution parameter must be chosen against an external functional criterion; module count is not a valid proxy for quality. Fourth, and most consequentially for practice, modularity does not robustly exceed degree-preserving null models on these networks (Section 3), and self-supervised embeddings only partially reproduce the inferred communities.

Implications for practice. Pathway-centric module analysis is common in transcriptomic studies, and the choice of partitioning method is rarely justified. Our results support three practical recommendations. (i) *Report functional coherence against a pathway-internal null.* Enrichment p-values are biased by module size and, within a functionally coherent pathway, are inflated for any partition; a size-matched, pathway-internal random baseline removes both biases. (ii) *Do not use module count or modularity alone as evidence of quality.* Modularity did not robustly exceed degree-preserving

null models, and module count is trivially controlled by the resolution parameter. (iii) *Select resolution against an external criterion* — functional coherence, stability across resamples, or downstream predictive performance — rather than by convention.

A robustness-oriented prescription. On clean data, multi-resolution partitioning is hard to improve upon: it already optimises a structural proxy well correlated with the functional target, so alternative mechanisms tend to trade one metric for another. The picture changes when the expression graph degrades — as it does in single-cell data, small clinical cohorts and noisy measurements. FACEO-R exploits that functional annotation is *unaffected* by expression degradation, and adapts its reliance on that annotation to the estimated reliability of the graph. The result is a method that does not sacrifice coherence on clean data while gaining markedly under degradation ($+0.046$ coherence under 50% feature dropout; 88% win rate; $p < 10^{-4}$). The practical value of annotation anchoring therefore lies not in beating structure-based methods on ideal data, but in preserving module quality when data are limiting — precisely the regime in which pathway-level module analysis is most often attempted.

Limitations. The analysis is restricted to nine GO biological-process pathways and sixteen cohorts; although this exceeds typical practice, it remains modest by the standards of large-scale benchmarking consortia. Density-based algorithms (DBSCAN, OPTICS, Mean Shift) were evaluated after spectral embedding to mitigate the curse of dimensionality, which may understate their performance on raw similarity profiles. Four high-complexity algorithms (edge betweenness, link communities, optimal modularity and full multi-scale Markov Stability) could not be evaluated reliably at the pathway sizes considered and were excluded; this is a limitation of feasibility rather than a negative finding about those methods. A further caveat concerns the attribution of the gain: the baselines operate on the expression graph alone, whereas FACEO-R additionally consumes GO-derived functional similarity, so part of its advantage reflects the availability of this additional information rather than the partitioning step itself. On the same anchored affinity, the relative ordering of partitioning algorithms changes, and establishing how much of the improvement is algorithmic rather than informational is an open question that a common-affinity benchmark would settle. We also note that the overlap expansion was retained for biological interpretability, since pleiotropy implies that a hard partition is an inappropriate model for gene function; on our benchmark it did not by itself improve functional coherence relative to a hard partition on the same affinity, and its benefit, if any, should be judged on biological rather than coherence grounds. Finally, the downstream evaluation is limited to disease-status classification, which is an easy task in these cohorts; survival and drug-target endpoints would be more discriminative.

Future work. Three directions follow naturally. First, extending the benchmark to single-cell and multi-omic data would test whether the conclusions hold when co-expression is estimated from sparser, more heterogeneous measurements. Second, overlapping community detection — included here only in a limited form — deserves systematic evaluation, since pleiotropy implies that hard partitions may be an inappropriate model for gene function. Third, replacing pathway-referenced benchmarks with experimentally derived complexes (e.g. CORUM) would provide a more direct biological ground truth.

5 Conclusion

On identical pathway-derived graphs, multi-resolution community detection produced functionally more coherent modules than hierarchical clustering, recovered nested sub-structure in synthetic benchmarks, and yielded better downstream prediction, while modularity did not robustly exceed degree-preserving null models. We therefore introduced FACEO-R, which anchors overlapping community detection on GO similarity through a reliability-adaptive weight and outperformed both Leiden and WGCNA-style partitioning (81% and 88% win rates) and on real single-cell data. We recommend that pathway-centric analyses report functional coherence against pathway-internal nulls and select resolution against external criteria rather than module counts or modularity.

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Competing interests

The authors declare no competing interests.

Data availability

All expression cohorts are publicly available from GEO under the accessions in Table 1. Processed expression matrices, complete evaluation tables (3100 main-benchmark, 528 synthetic-benchmark, 317 downstream, 435 FACEO-R degradation and 60 single-cell validation records), statistical outputs, all figure-generating scripts and the manuscript source are organised as a reproducible pipeline and are available at https://github.com/sunmy1129/faceo-r. Single-cell data: GEO GSE145926, GSE185224.

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